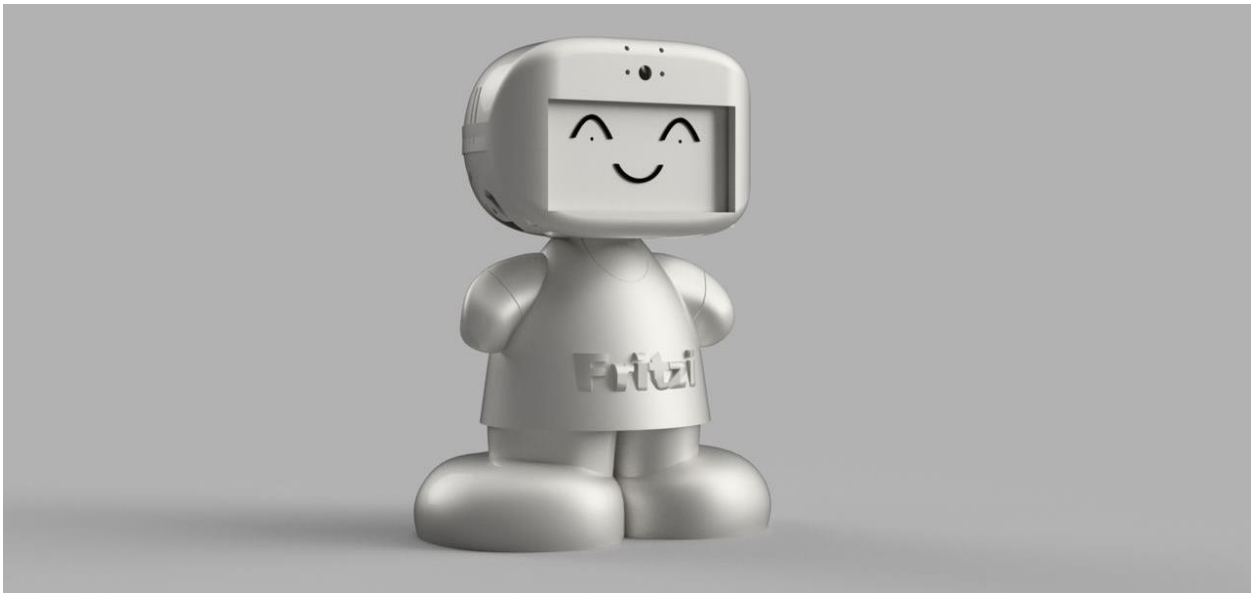
Final Report / Project B

Conversational Social Voice Assistant for Nursing Home Residents



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Introduction

As our global population undergoes a profound demographic shift towards an increasingly aging society, marked by the "longevity revolution," the imperative to design inclusive and effective products for older adults becomes paramount. This revolution is propelled by more stable political situations worldwide, advances in medicine and technology, and improved access to healthcare, resulting in increased lifespans and lower mortality rates. However, these advancements also usher in unique challenges that necessitate innovative solutions.

Global population projections illustrated in Figure 1, conducted by the UN Population Division, underscore the diverse trajectories the world may experience, ranging from continuous growth to more moderate scenarios that stabilize or even decline. Regardless of the specific trajectory, the UN anticipates a continued global population increase until at least 2050.

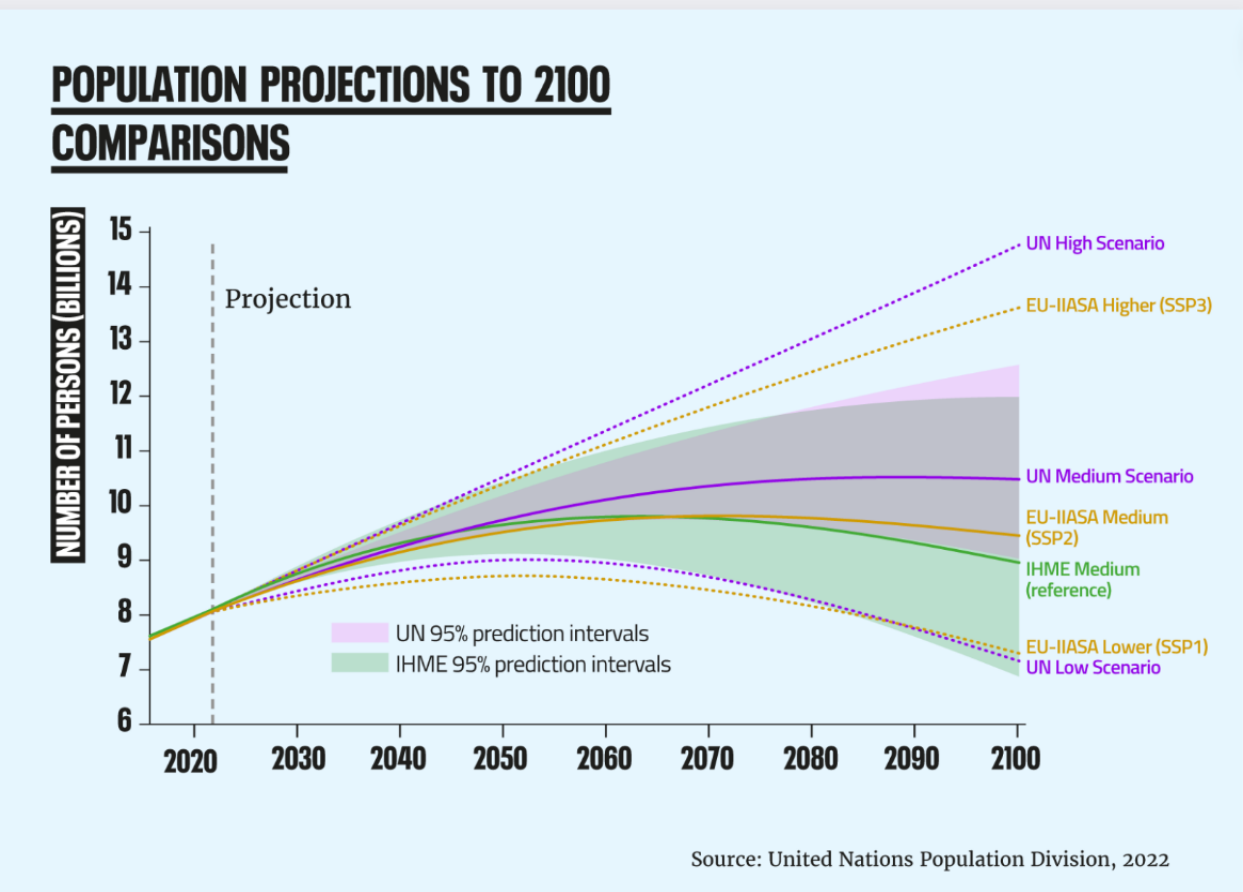


Figure 1. Comparison of Different Global Population Projections to 2100

Figure 2 visually represents these projections, indicating that in 2020, about 9% of the global population, or approximately 1 in 11 people, was aged 65 or older. By 2050, this percentage is expected to rise to 16%, equating to approximately 1 in 6 people falling into this age category.

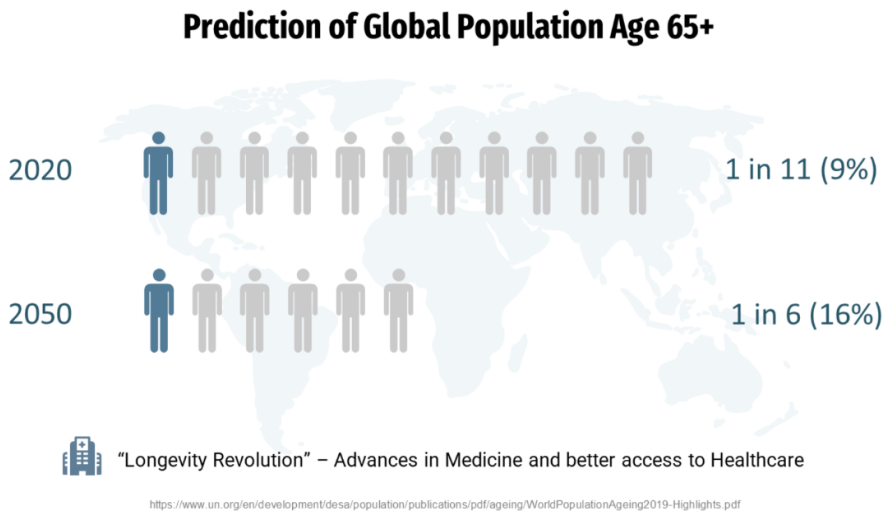


Figure 2. Prediction of Global Population Age 65+

This demographic shift is not confined to the global scale, as Figure 3 showcases a localized perspective for Germany. In 2020, about 22% of the German population, or every 1 in 5 people, was already aged 65 or older. Projections indicate that by 2050, this figure is expected to increase to around 30%, meaning roughly every third person in Germany will be aged 65 or older.

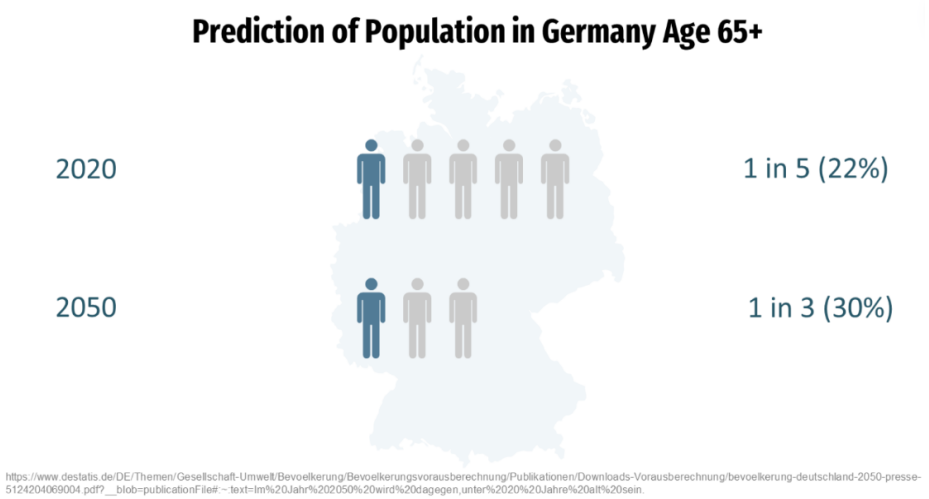


Figure 3. Prediction of Population in Germany Age 65+

The challenges associated with the "longevity revolution" are further illustrated in Figure 4, which depicts the population pyramid of Germany in 2005 alongside a projected distribution for 2050. The left stacked histogram reveals a bulge in the middle section between the ages of 30 and 60, representing the "Baby Boomer" generation resulting from previous high birth rates. These individuals, a driving economic force in their prime years, are anticipated to retire from the workforce en masse until 2030. The right-hand histogram in Figure 4 projects this demographic transition into 2050, illustrating a more constrictive pyramid that widens at the top and middle sections but narrows at the bottom, indicative of a declining population.

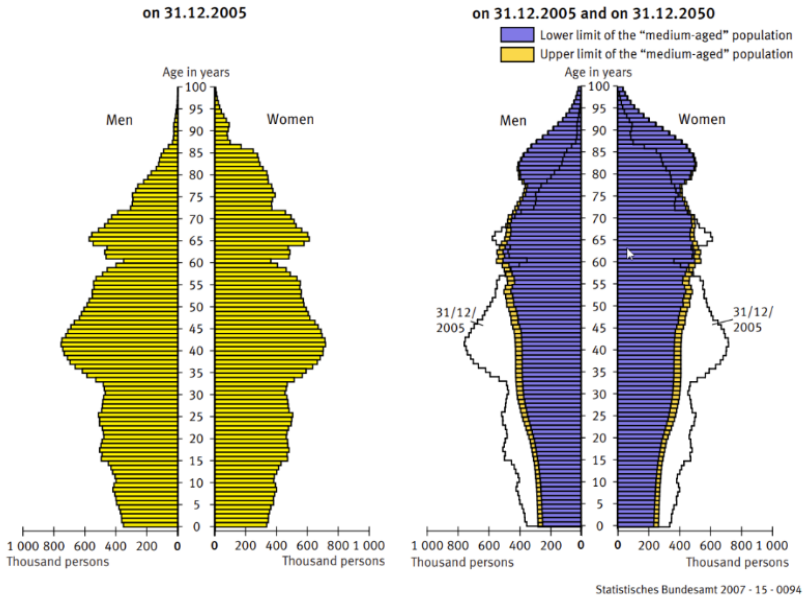


Figure 4. Population Pyramid for Germany in 2005 (Left) and Projection for 2050 (Right)

Figure 5 offers a concrete example of this social development, highlighting the increasing development and rise in mobile care services and nursing homes in Germany. Between 1999 and 2021, their numbers have almost doubled, and this trend is anticipated to persist in the future. However, a pressing challenge emerges, both evident today and exacerbated by the projected developments in Figure 4 – the shortage of skilled workers. As life expectancy increases and the birth rate decreases, the distribution of the population shifts primarily towards adults and older adults. Consequently, the gaps in the labor market left behind by retiring older adults cannot be feasibly filled by younger generations. In the context of our study, this means that, in the future, a growing number of older adults will require nursing services and facilities to assist them in leading dignified lives, while simultaneously fewer and fewer young people will be available to work in these crucial sectors.

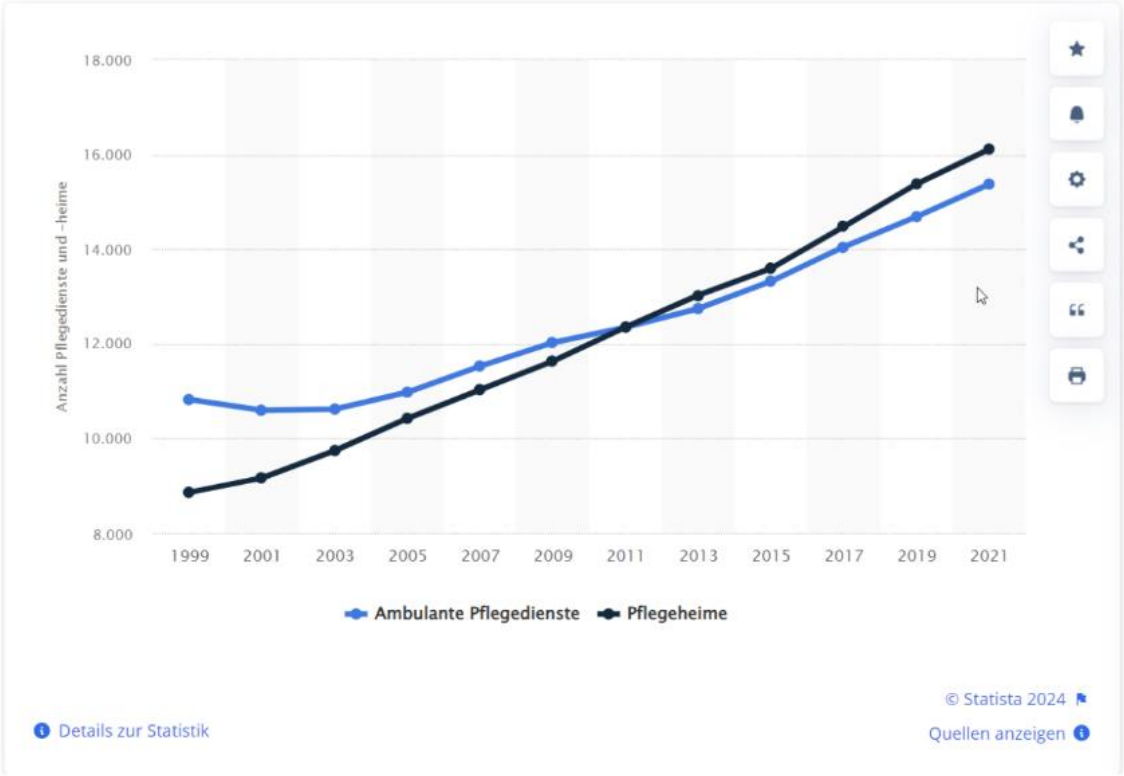


Figure 5. Development Numbers of Mobile Nursing Services and Care Homes in Germany 1999-2021

With a considerable portion of the workforce transitioning into retirement, the need for possible technological solutions catering to the unique requirements of older adults becomes increasingly relevant. This research project focuses on addressing this need by exploring the potential of a conversational voice assistant as an HCAI solution within the specific context of a retirement home in Germany.

Motivation behind our idea

The genesis of our project stems from a profoundly relatable sentiment expressed by a participant in a research paper led by Prof. Müller et al.:

> *"I have a radio, but with these buttons, it is too complicated. That's why I do not use it. […] I would like to listen to it if it were simpler"* (Müller et al., 2012, p. 2641).

This poignant statement encapsulates the struggles many older adults face when attempting to engage with technology designed without due consideration for their unique needs and capabilities.

The participant's frustration with the complexity of a seemingly straightforward device, the radio, served as a stark reminder of the barriers that hinder older individuals from accessing the benefits of modern technology. Recognizing the potential of technology to enhance the lives of older adults, our team was motivated to bridge this gap and create a solution that addresses the specific challenges faced by this demographic.

Recent advancements in AI technologies, particularly in Language Models (LLMs), have opened new possibilities for human-computer interaction. The advent of Speech-to-Text (STT) technology allows for more accurate transcription of spoken language into text, liberating users from traditional command-based approaches. Natural Language Processing (NLP) advancements enable systems to process human language more accurately, promoting context-aware and conversational interactions. Text-to-Speech (TTS) technologies have evolved to synthesize more realistic and natural-sounding utterances, enhancing the overall user experience.

Our objective crystallized around developing a conversational voice assistant, leveraging these AI advancements, to provide a more intuitive and accessible interface for older adults. By harnessing the capabilities of commercially available off-the-shelf STT, NLP, and TTS solutions, we aimed to create and test a prototype that makes technology more accessible and intuitive for users across diverse demographics. The desire expressed in the participant's plea for simplicity in using a radio inspired us to design a solution that not only simplifies interactions but also fosters independence and engagement among older users.

Challenges faced by older adults in socializing, as highlighted by Müller et al. (2012), further underscore the importance of our project. Social interaction for older adults becomes complex due to factors such as family unavailability, behavioral constraints in their living environments, and shortages of staff and event offers in care facilities. Studies, such as those conducted by Koszalinski & Olmos (2022), emphasize the positive health benefits, both physical and mental, that communication can bring to older adults. As demonstrated by Nakagomi et al. (2022), even digitally mediated communication seems to foster these health benefits. Through this project, we aim to contribute to the

broader discourse on human-computer interaction and the imperative of addressing the unique challenges posed by the "longevity revolution" by investigating the following questions:

- Can an AI-based social companion fill the shortage of the skilled labor gap and become a (surrogate) conversation partner for older adults?
- If and how will it be appropriated by older adults?

The following transcript from one of our conducted workshops at the nursing home left us optimistic.

> *[CHRIS]: So, summarizing again, there could be advantages, due to a shortage of skilled workers and so on, for example?*
>
> *[MR. X]: Yes, definitely!*
>
> *[CHRIS]: You would prefer people, but if no staff can be found, something like that would still be...?*
>
> *[MR. X]: Then I would be stupid if I didn't take advantage of this opportunity if it is given to me.*

Research problem within the context of HCI

The landscape of Human-Computer Interaction (HCI) is rapidly evolving, propelled by demographic shifts, technological advancements, and a growing awareness of the unique challenges faced by specific user demographics. The intersection of these factors has given rise to a compelling research problem that demands attention within the context of HCI.

1. Demographic Shifts and Aging Population:
The global demographic transition towards an increasingly aging society, often termed the "longevity revolution," poses a multifaceted challenge. Figures 1, 2, and 3 illustrate the anticipated increase in the proportion of older adults, particularly in countries like Germany. This demographic shift not only necessitates a reevaluation of existing

technology but also demands the development of new HCI solutions that cater to the needs and preferences of older users.

2. Barriers to Technology Adoption for Older Adults:
The motivational spark for our project originated from a participant's frustration with the complexity of a radio, symbolizing the broader issue of barriers faced by older adults in adopting and interacting with technology. The research problem lies in the insufficient consideration of the unique needs and abilities of older users in the design and functionality of digital systems, leading to a digital divide that impacts their quality of life.

3. Social Isolation and Communication Challenges:
The complexities of socializing for older adults, as highlighted by Müller et al. (2012), amplify the importance of HCI solutions that facilitate communication and connection. This extends beyond simplifying technology interactions to addressing the challenges of social interaction faced by older individuals due to factors such as family unavailability, behavioral constraints, and staff shortages in care facilities.

4. Technological Advances and Conversational Interfaces:
Recent advancements in AI technologies, including Speech-to-Text (STT), Natural Language Processing (NLP), and Text-to-Speech (TTS), provide an opportunity to reshape the HCI landscape. The challenge lies in effectively harnessing these technologies to create user-friendly, intuitive, and accessible interfaces, particularly through conversational voice assistants that cater to the needs of older adults.

5. Health Benefits of Communication:
Studies like Koszalinski & Olmos (2022) underscore the positive health benefits of communication for older adults, both physically and mentally. The research problem within HCI is to explore and implement solutions that leverage technology to enhance communication, combat social isolation, and contribute to the overall well-being of older individuals.

In addressing these research problems, our project focuses on developing a conversational voice assistant tailored to the unique needs of older adults in a retirement home in Germany. By combining insights from demographic trends, technological advancements, and the challenges highlighted in HCI research, we aim to contribute to

the ongoing discourse and advancements in creating inclusive, accessible, and empowering technology solutions for an aging population.

Objectives of the study

This research endeavors to explore the viability of an AI-based social companion as a surrogate conversation partner for older adults and its potential in mitigating the shortage of skilled labor, particularly in settings like nursing homes. The study seeks to understand the appropriation of this technology by older adults, examining if and how they interact with our prototype, its usability, and perceived benefits or challenges. Employing a Research through Design (RtD) methodology, we aim to design, develop, and evaluate the AI-based social companion prototype, ensuring a user-centered approach. The study further includes a Co-Design Workshop involving older adults to actively engage them in the design process and a Focus Group with nursing home residents to gather insights into their experiences, perceptions, and expectations regarding AI technology. Through these objectives, we aim to contribute valuable insights into the intersection of AI technology, human-computer interaction, and the specific needs of older adults, fostering a deeper understanding of the potential benefits and challenges associated with the adoption of AI-based companions in care contexts.

Literature Review

Participatory Design with older adults

The literature on participatory design with older adults underscores the growing interest in incorporating their perspectives to enhance the design of digital technologies tailored to their needs. Davidson and Jensen (2013) advocate for the involvement of older adults in technology ideation, emphasizing the benefits despite potential hindrances like critiquing existing applications. Similarly, Johnson and Finn (2017) emphasize the imperative of improving digital user experiences for older adults, acknowledging the contributions of various stakeholders. Pradhan et al. (2020a) delve into the efficacy of design workshops in engaging older adults, highlighting the importance of designing for individual needs rather than generalized assumptions. Orso et al. (2015) exemplify the

adaptation of participatory design methods to meet older users' specific needs through the development of an interactive television prototype. Likewise, Coventry et al. (2012) introduce the "technology tea party" methodology as a means to involve older adults throughout the design process, emphasizing early engagement and requirement elicitation. Marzi et al. (2023) further investigate the motivations and preferences of older adults in participating in HCI research, advocating for less commitment and more open formats in participatory design. Meanwhile, Duque et al.'s systematic review (2019) underscores the necessity to enhance traditional design approaches to better accommodate the needs of elderly users. Additionally, Correia et al. (2020) demonstrate the use of illustration to create more inclusive interfaces for older adults, emphasizing the significance of building strong relationships with older users. Loi and her colleague (2020) examine how stereotypes influence product design and HCI, emphasizing the importance of centering design on the lifeworlds of technology users. In a related vein, Lindsay et al. (2012) propose participatory design as a solution to improve the design quality of digital technologies for older adults, outlining an approach to early engagement and discussing challenges encountered in various design projects.

Through these studies, the literature highlights the significance of actively involving older adults in the design process to create more inclusive and effective digital solutions for this demographic. Building upon the insights gleaned from these studies, we utilized their findings as foundational guidance for planning and executing our research workshops with older adults, thereby fostering a collaborative environment conducive to the creation of tailored and inclusive digital solutions.

(Conversational) Voice Assistants and older adults

The integration of Conversational Voice Assistants (VAs) into the fabric of daily life holds significant promise for augmenting the lives of older adults, a group traditionally marginalized in the design and application of such technologies. Scholarly research, exemplified by the works of Dosso et al. (2023), Ringfort-Felner et al. (2022), and Sidner et al. (2018), places a strong emphasis on involving end-users in discussions that delve into the elements influencing emotional connections with social robots and intelligent personal assistants (IPAs). These discussions underscore the pivotal role of user engagement in mitigating barriers to technology adoption among the elderly and those living with dementia, highlighting the necessity for designs that are both personalized and empathetic.

Further compounding this discourse, Sarcar et al. (2023) and Seiderer et al. (2020) champion the cause for inclusivity within the realm of Conversational User Interface (CUI) design targeting older adults. They address the demographic's underrepresentation in voice and chat-based interfaces, proposing solutions that span interdisciplinary collaboration and adhere to privacy-by-design principles. By establishing forums for discourse among researchers, designers, practitioners, and end-users, their efforts are instrumental in fostering a culture of inclusivity in CUI design tailored to the aging population.

Additionally, the contributions of Razavi et al. (2022), Sin et al. (2022), and Lines and Hone (2002) to the field merit recognition. Their research investigates the utility of conversational agents, voice user interfaces (VUIs), and speech output in catering to the social, informational, and healthcare requisites of older adults. A significant aspect of their findings revolves around bridging the dichotomy between user expectations and actual experiences, a crucial step towards fostering acceptance and satisfaction. Insights into the impact of response delay on user satisfaction, as explored by Peng et al. (2020), are invaluable for devising strategies to counteract negative perceptions during interactions with conversational robots.

Moreover, the investigations undertaken by Pradhan et al. (2020b) and Oewel et al. (2023) shed light on older adults' perceptions and utilization of voice assistants in the context of health-related and social well-being objectives. These studies elucidate the technology's potential in meeting the distinctive needs and preferences of senior populations, thereby informing the development of voice assistants that bolster aging in place and improve life quality.

Echoing this sentiment, Sayago et al. (2019) and Cuadra et al. (2023b) call for an expanded exploration of voice assistants designed with and for the elderly. They pinpoint critical gaps in voice interface design and the prospects of voice-first ambient interfaces in supporting seamless aging in place. By prioritizing the unique needs and preferences of older adults, their research endeavors to advance digital inclusion and empower this demographic to leverage voice-enabled technologies.

The inquiries of Upadhyay et al. (2023), and Trajkova and Martin-Hammond (2020), along with the analyses provided by Brewer (2022), Cambre and Kulkarni (2019), and Clark et al. (2019), delve into the dynamics of older adults' adoption and discontinuation of intelligent voice assistants. They examine perceptions, social implications, and design flaws of AI-powered voice technologies, advocating for user-centered design paradigms

and a reevaluation of design norms to elevate user experience and promote digital inclusivity among older adults.

Coghlan et al. (2022) compare older adults' preferences for accessing information via touchscreens versus voice-user interfaces, while Chen et al. (2023) investigate the standalone benefits of combined voice and touchscreen Voice Assistants for this demographic. Desai and Chin (2023) introduce an innovative application, Health Buddy, embedded within Voice User Interfaces (VUIs) to facilitate informal self-regulated learning on health topics, underscoring the implications for designing VUI-based educational programs. Basson et al. (2007) and Cuadra et al. (2023a) underscore the critical need for voice assistant designs that are mindful of older adults' requirements, advocating for digital inclusivity through multi-modal interactions in varied settings.

Collectively, these studies illuminate the path forward in the design, usability, and transformative potential of voice assistants on the lives of older adults. They assert the indispensability of user-centered design strategies that meticulously cater to the specific needs and preferences of this demographic, thereby endorsing digital inclusion and elevating the quality of life.

After reviewing these papers, we have identified a gap in the research literature regarding the utilization of Large Language Models for voice-based conversational purposes among older adults. While the studies reviewed offer valuable insights into the design, usability, and transformative potential of voice assistants, there is a relative dearth of research exploring the integration of advanced language models in conversational interfaces tailored to the needs of the aging population. As such, our project aims to address this gap by leveraging the capabilities of Large Language Models to enhance the conversational experiences of older adults, thereby contributing to the advancement of digital inclusivity and quality of life for this demographic. While recent projects like "Navel - the social robot", delve into similar aspects, our focus on harnessing language models for conversational purposes presents a novel contribution to the field, with the potential to complement existing efforts in enriching the interaction between older adults and technology.

Methodology

Research Through Design

This project adopts a Research Through Design (RtD) methodology, influenced by the works of Zimmerman, Forlizzi, and Evenson (2007), who advocate for RtD as a means to create innovative technologies and generate new understandings through design practice. Compared to "Research for Design" and "Research about Design," Research Through Design (RtD) places a distinct emphasis on the process of design itself as a primary mode of inquiry. Unlike "Research for Design," which focuses on gathering information to inform the design process, and "Research about Design," which examines the outcomes and implications of design activities, RtD actively engages in the creation of artifacts to explore and address research questions, thereby generating new insights and understanding through the act of designing.
Our focus is on developing a conversational social voice assistant tailored to the needs and preferences of nursing home residents. The guiding research questions are:

- Can an AI-based social companion fill the gap and become a (surrogate) conversation partner for older adults?
- And, (how) will it be appropriated by older adults?

These questions draw on the concerns raised by Rogers and Marsden (2013) regarding the adoption and appropriation of technology by older adults, emphasizing the need for designs that cater to their unique contexts and experiences.

RtD Process

Our RtD process consists of the following phases:

Problem Identification: Initially, we conducted desk research and a comprehensive literature review to explore the social needs and technological competencies of older adults. This approach was informed by the synthesis of existing knowledge and insights gathered from relevant studies in the field of aging and technology.

Conceptualization, Co-design and Focus Group: Drawing from Sanders and Stappers (2008) on co-creation, we engaged older adults in the design process to ideate and

conceptualize potential features and interactions with the digital assistant. This phase ensures that the design is grounded in the actual needs and desires of the target users.

Prototyping and Iteration: Based on these insights, we developed a prototype, ranging from initial low-fidelity mockups to a functional 3D printed voice assistant, to explore various interaction modalities. Buxton (2007) emphasizes the value of sketching and prototyping in design as tools for exploration and communication.

Evaluation: Our evaluation phase involved testing and analyzing the first version of our prototype over a span of 10 days at a senior center. During this period, we captured participants' interactions with our prototype “Fritzi” through voice and video recordings and data logs. These were later analyzed to assess the level of engagement, interest, and ease of interaction with the prototype. Detailed results of this evaluation are presented in the Results and Discussion section.

Reflective Analysis: The project concludes with a reflective thematic analysis aimed at addressing our research questions: Can an AI-based social companion fill the gap and become a (surrogate) conversation partner for older adults? And, (how) will it be appropriated by older adults? We draw on the reflective practice model of Schön (1983) to critically analyze our design process, outcomes, and the implications for HCI and aging, focusing on the insights gained in relation to these research inquiries.

Contribution to Knowledge

Our RtD approach contributes to the knowledge base in several ways. Firstly, it provides empirical evidence on the feasibility and acceptance of AI-based social companions by older adults. Secondly, it offers insights into the design and implementation of voice-assisted technologies that are sensitive to the capabilities and preferences of older users. Lastly, this project highlights the potential of RtD in addressing complex, real-world problems by closely involving end-users in the design process, thereby ensuring that the outcomes are both innovative and grounded in user needs.

Participant Recruitment Process

Initially, our access to potential participants was facilitated through a collaboration with a senior center, facilitated by Professor Müller, a respected expert in gerontology research with whom we have an established relationship.

The recruitment of participants for both the Co-design workshop and focus group sessions followed a structured and professional approach to ensure the involvement of suitable and engaged individuals. We engaged in multiple meetings with the management of the senior center to present our research objectives and methodology, seeking their approval and cooperation in hosting our workshops. After thorough deliberation, the senior center management granted us permission to conduct our activities within their facilities, recognizing the value of our research in enhancing the well-being of their residents.

With the endorsement of the senior center, we were able to create a conducive environment for our workshops, ensuring that participants felt comfortable and supported throughout the process. The management of the senior center facilitated the recruitment process by providing us access to a pool of interested and eligible participants who met the criteria for inclusion in our study. The sampling process utilized Convenience and Purposive sampling strategies, ensuring that participants were selected based on their availability and suitability for the study objectives.

Throughout the recruitment process, we prioritized transparency and ethical considerations, ensuring that participants were fully informed about the nature and objectives of the study, as well as their rights and responsibilities as research participants. We obtained informed consent from all participants prior to their involvement in any research activities, emphasizing their voluntary participation and the confidentiality of their responses.

Despite the successful recruitment of participants, we encountered several challenges during the workshops, which will be discussed in detail in the subsequent sections. These challenges provided valuable insights into the complexities of engaging older adults in research activities and informed our approach to data collection and analysis.

Data Collection Methods

Co-Design workshop

The data collection for this study was centered around the implementation of a co-design workshop methodology. This approach is grounded in participatory design principles, which emphasize the direct involvement of stakeholders, including users, in the design process. The co-design workshop facilitates a collaborative environment where participants can contribute their perspectives, experiences, and ideas towards the development of a solution or product (Sanders & Stappers, 2008).

This co-design workshop methodology aligns with the recommendations of previous studies, which have highlighted the effectiveness of participatory approaches in generating rich, contextual, and user-centered data (Brandt et al., 2012). By actively involving end users in the design process, this method not only aids in the collection of valuable data but also contributes to the development of solutions that are more likely to meet user needs and expectations. In the subsequent section, we thoroughly examine the proceedings and results of our co design workshop, thereby deepening our comprehension of the perspectives and preferences held by older adults.

Conducting Focus Group

This research employed a focus group as a pivotal data collection technique, leveraging the dynamic interplay of opinions, experiences, and attitudes among participants to delve into the study's themes. Focus groups are a qualitative research method that assembles a small, diverse group of individuals to discuss specific topics of interest in a structured setting. This method is particularly valued for its ability to generate a rich understanding of participants' perspectives and experiences through the facilitation of an interactive group process (Robinson, 2020).

The employment of focus groups in this study is supported by literature that highlights their effectiveness in exploring complex behaviors, perceptions, and motivations among groups of individuals (Stewart & Shamdasani, 2017). By enabling direct interaction among participants, focus groups can uncover deeper insights and foster a richer understanding of the subject matter, making them an invaluable tool for qualitative research. In the next section, we delve into the process and outcomes of our focus group discussions, further enriching our understanding of older adults' perspectives and preferences.

Qualitative Data Analysis

Throughout the project, we gathered a diverse array of data, providing valuable insights into user interactions and experiences. This comprehensive dataset encompassed field notes taken during our visits and workshops at the nursing home, offering a deeper contextual understanding of the environment and user interactions. Additionally, informal discussions with staff members and residents yielded further perspectives and nuanced insights into their needs, preferences, and challenges. The log files generated by the prototype captured system interactions, errors, and user actions, furnishing us with some quantitative metrics and technical insights. Audio recordings of user interactions with the prototype, along with accompanying transcripts, facilitated qualitative analysis of conversation flows, topics, and user engagement. Moreover, video recordings of users interacting with the prototype enabled visual analysis of user types and engagement patterns. Open feedback forms provided to users enabled the collection of their impressions, suggestions, and concerns, furnishing valuable qualitative data on user experiences and preferences.

This multifaceted dataset formed the foundation for our qualitative data analysis, allowing us to explore user needs, preferences, and experiences with the prototype. We employed thematic analysis to delve deeper into the specific topics that users engaged with while interacting with our prototype, identifying recurring themes and patterns. Through thorough analysis and interpretation of this data, we gained valuable insights to guide further refinement and enhancement of the prototype.

User-Centric Design in Action: Prototyping and Testing a Voice Assistant for Older Adults

Selection of hardware and software

Hardware

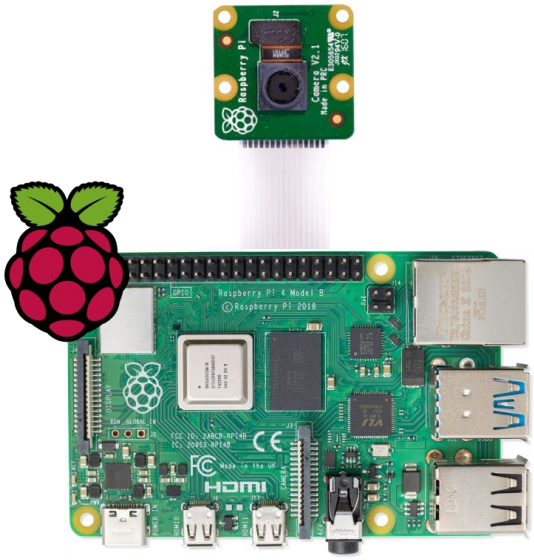
In the process of determining the most suitable hardware for our conversational voice assistant project, we engaged in an iterative testing phase to ensure optimal

performance. Initially, the concept involved running the entire software locally, including Speech-to-Text (STT), Language Models (LLM), and Text-to-Speech (TTS). To evaluate this setup, we acquired a donated desktop computer (Fujitsu Esprimo P757/E85+ with an Intel Core i5 and 4GB RAM) from the FabLab, intending to use it as a server. Raspberry Pi Zero devices were considered as "Satellites" to record user utterances, transmit them to the server for processing, and receive the synthesized voice output.

However, tests revealed that the desktop computer lacked the necessary processing power, particularly without a dedicated GPU. The Real-Time Factor (RTF) for achieving a decent-sounding German voice model exceeded 30, indicating that the computer would require 30 seconds for every second of synthesized speech output. This impractical delay, especially for longer responses, led us to reconsider our hardware configuration.

Aligned with the project's commitment to showcasing a prototype that utilizes commercially available off-the-shelf products, and considering budget constraints as we funded the components independently, we opted for an alternative hardware setup:

As the foundation for our conversational voice assistant prototype, we strategically chose the Raspberry Pi 4 Model B (8GB RAM) paired with a Raspberry Pi Camera Module v2. The decision to leverage this hardware was driven by its compact form factor, coupled with a sufficiently capable CPU and ample RAM, instilling confidence in our ability to



create a functional and efficient prototype. The Raspberry Pi's versatility and community support made it an ideal choice for our project.

Recognizing the challenges identified in previous tests with self-hosted software solutions, where running all processes locally proved to be nearly impossible, we adopted a pragmatic approach. Instead of hosting the entire software stack locally, we decided to integrate commercially available software solutions for Speech-to-Text (STT), Natural Language Processing (NLP), and Text-to-Speech (TTS) components (see Software). This strategic shift allowed us to capitalize on the capabilities of the Raspberry Pi hardware while utilizing external services for computationally intensive tasks, ensuring a balance between functionality and computational efficiency.

The Raspberry Pi's small footprint, coupled with its respectable computational capabilities, aligns with our objective to create a prototype that is not only effective but also practical and accessible, embodying the spirit of using off-the-shelf components for real-world applications.

To enhance the storage capabilities of our conversational voice assistant prototype, we repurposed a Samsung Evo 870 SSD SATA III (500GB) initially acquired for testing on the desktop computer. Recognizing the unique demands of our project and the need for reliable, high-performance storage, we opted to utilize the SSD in conjunction with the Raspberry Pi.

While the Raspberry Pi can operate with an SD card as its primary storage, the decision to integrate an SSD was driven by several considerations. Sequential reads and writes are notably more reliable and faster on an SSD, providing a performance boost that proves beneficial for our application. Moreover, SSDs offer increased durability and longevity compared to traditional SD cards, making them more resilient in write-heavy environments. By leveraging the advantages of SSD technology, we aim to ensure seamless operation and enhance the overall performance of our conversational voice assistant prototype.

To address the audio in- and output requirements of our conversational voice assistant prototype, we selected the Waveshare WM8960 Audio HAT with Stereo Speakers. This HAT (Hardware Attached on Top) can be attached directly to the GPIO pin headers of the Raspberry Pi. The decision to integrate this audio HAT was motivated by several key features that enhance the functionality and flexibility of our prototype.

The Waveshare WM8960 Audio HAT offers dual microphones, enabling effective audio input capture. Additionally, it provides connectivity for stereo speakers, contributing to a rich and immersive audio output experience for users interacting with the voice assistant. It also includes a dedicated programmable button that adds a valuable element for interactions with the prototype (see Interaction model of the prototype). The small form factor of the Waveshare WM8960 Audio HAT aligns with our objective to maintain flexibility in the exterior design of the finished prototype.

As our project evolved and following the insightful workshops conducted with residents of the nursing home, the design of our conversational voice assistant prototype took on a more tangible form. Initially considering a box or cylinder, the feedback from the workshops guided us towards a more anthropomorphic design, resembling a small robot. This shift in design was driven by the desire to create a device that not only houses the technology but also fosters a sense of companionship through its physical form.

However, this design evolution posed a challenge in terms of accessibility to the programmable button on the Waveshare WM8960 Audio HAT. The wiring and component placement made it impractical to reach the button directly. To overcome this limitation, we incorporated an inexpensive LogiLink USB Numpad into the design.

Through a software addon and a shell script, we were able to reprogram certain buttons on the Numpad to trigger essential functions of the voice assistant. This included initiating the listening function and adjusting the volume of the stereo speakers.

Software

For the operating system (OS) to run on the Raspberry Pi, we opted for Raspberry Pi OS Lite, released on December 11th, 2023. This OS version is designed for 64-bit systems and utilizes Kernel version 6.1, with Debian version 12 (bookworm).

The decision to choose Raspberry Pi OS Lite was influenced by several factors. Firstly, the Lite version was preferred due to the absence of a display attached to the Raspberry Pi. This allowed us to conserve resources that would otherwise be allocated to support a graphical desktop environment. By selecting the Lite version, we optimized the performance of the Raspberry Pi for our specific use case, prioritizing efficiency and functionality.

Furthermore, the configuration process was streamlined by connecting to the Raspberry Pi via SSH and making necessary changes through the command-line interface (CLI). This approach provided flexibility and convenience in managing the Raspberry Pi remotely, eliminating the need for a physical connection to a display or peripherals.

motionEye
(Motion detection)

In response to a last-minute suggestion from our Professor to record user interactions with our prototype on video, we integrated motionEye Version 0.42.1 into our system. motionEye provides an online interface for the software *motion*, a video surveillance program with motion detection capabilities.

The decision to incorporate motionEye was motivated by the need to capture user interactions with our conversational voice assistant prototype in a non-intrusive manner. We wanted to ensure that only individuals engaging with our prototype would be recorded, thus preserving privacy and ethical considerations. motionEye facilitated this by allowing us to create "masks" within the video feed, highlighting specific areas where motion detection would trigger recording.

While we were unable to fully configure the software to implement the masking feature due to time constraints, the recorded videos still provided valuable insights. The general motion detection and recording functionality allowed us to capture user interactions and assess the level of engagement with our prototype, the number of users involved in interactions and their interaction patterns.

At the heart of our conversational voice assistant prototype lies OpenVoiceOS (OVOS), a community-driven, open-source voice AI platform tailored for creating custom voice-controlled interfaces across various devices. OVOS boasts features including Natural Language Processing (NLP), a customizable user interface, and a strong emphasis on privacy and security.

The selection of OVOS as the core software for our prototype was informed by several factors. Firstly, some team members had prior experience with its predecessor, Mycroft AI, which instilled confidence in its capabilities and reliability. Additionally, the open-source nature of OVOS appealed to our ethos of transparency and collaboration, enabling us to tailor the software to our specific requirements.

OVOS's compatibility with a wide range of devices and its high level of customizability were also significant factors in our decision-making process. These qualities allowed us to seamlessly integrate OVOS into our hardware setup and tailor its functionalities to meet the unique needs of our conversational voice assistant prototype.

Furthermore, OVOS's active and supportive community provided valuable resources and assistance throughout our prototyping process. Despite the project being managed by a small team, the active community ensured access to support and guidance whenever needed.

While OVOS offers the capability to run fully offline, our tests indicated a necessity to rely on external services, particularly for Speech-to-Text (STT), NLP, and Text-to-Speech (TTS) functionalities. OVOS's extensive library of plugins enabled us to explore and test different external services to determine the best fit for our project before making a final decision.

In our effort to optimize our conversational voice assistant prototype, we sought external services for critical functions like Speech-to-Text (STT) and Text-to-Speech (TTS). After

thorough evaluation, we opted to integrate Microsoft Azure, Microsoft's cloud-computing platform, for these services.

Our decision was guided by several factors. During tests while working on the prototyp, we found Microsoft Azure provided very accurate transcriptions of spoken utterances, crucial for ensuring effective communication, especially considering the diverse speech patterns of older adults. Moreover, Azure offered a wide selection of high-quality German voice models, enhancing the user experience by minimizing the robotic sound often associated with synthesized speech.

Additionally, Azure demonstrated low latencies for both transcription and speech synthesis, ensuring responsiveness and smooth interactions with the voice assistant. This was vital for maintaining conversation flow and reducing user frustration.

At the time of prototype development, access to Large Language Models (LLMs) was limited. While various options have since become available, during our development phase, only OpenAI's model was accessible via a plugin through OVOS. However, leveraging the GPT-4 Model API, we were able to harness its powerful Natural Language Processing (NLP) capabilities to facilitate conversational interactions within our prototype.

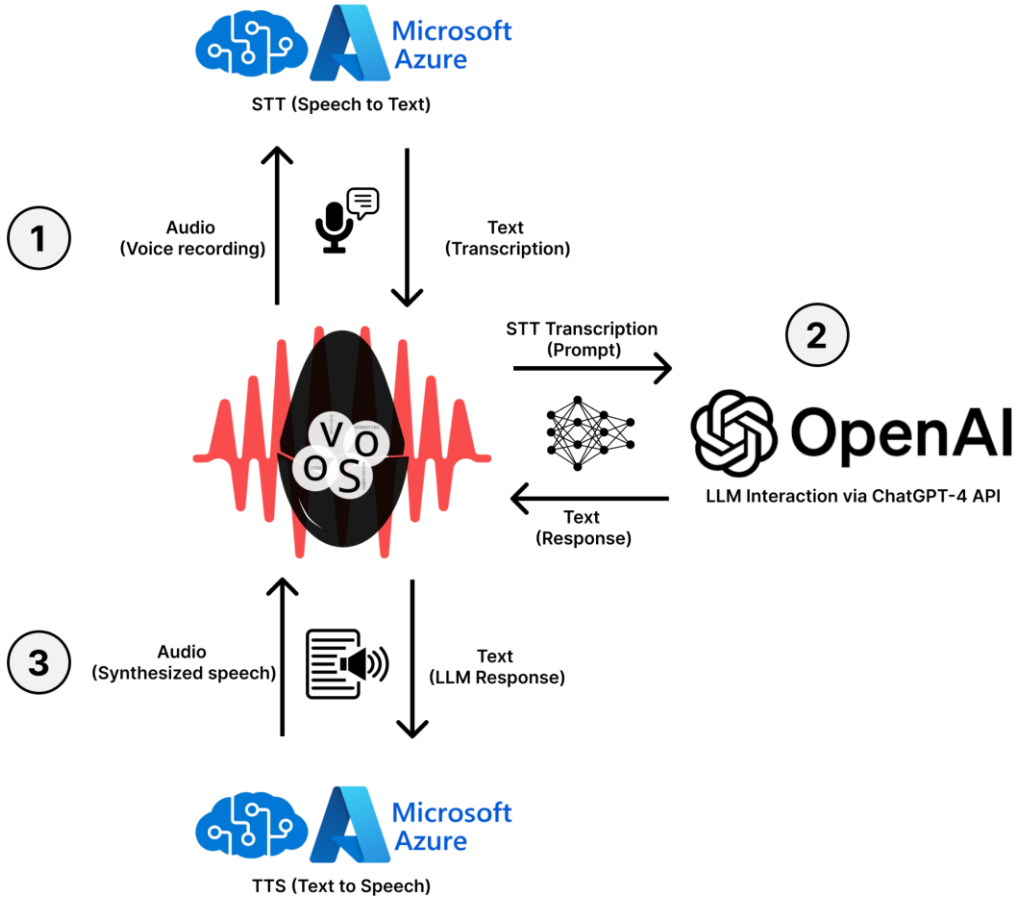


Figure 6. Data Processing Flow for our Prototype

The diagram in Figure 6 illustrates the data processing flow of our prototype:

1. The user's utterance is recorded and sent to Microsoft Azure's Speech-to-Text (STT) service. The transcription is then returned to OVOS.

2. The transcript, along with a "persona[1]" prompt defined in the ChatGPT plugin settings in OVOS, is sent to ChatGPT-4 API. The persona provides context for the Large Language

[1] Persona Prompt: "Als freundlicher digitaler Assistent im Altersheim: Beantworte Bewohnerfragen klar und kurz und spreche sie per Du an. Zeig Interesse an ihren persönlichen Geschichten, stelle gezielte Rückfragen und ermutige zu weiteren, um ein tieferes Verständnis zu fördern. Vermeide Aufzählungen, Abkürzungen und verwende einfache Sätze für maximale Klarheit auf persönlicher Ebene."

Model (LLM) on how to respond to the transcript. The response from ChatGPT is then sent back to OVOS.

3. OVOS sends the response from the LLM to Microsoft Azure for Text-to-Speech (TTS) synthesis. The synthesized speech of this response is played through the speakers to the user.

Following the output from our prototype, users have the option to provide a follow-up question or response. This input is directed back to the prototype, initiating the sequence once again. The process continues iteratively, allowing for a seamless and dynamic conversational interaction between the user and the prototype.

Co-design workshop with older adults in senior home

Our project team organized a co-design workshop, Figure 6, initially intended as a participatory design session, aimed at understanding the preferences of the elderly concerning the visual and interactional aspects of voice assistant technology. The goal was to engage directly with this demographic to gather insightful feedback and incorporate their preferences into the design process. We initially envisioned incorporating participatory design principles into our project. However, upon realizing the limited number of interested and willing residents, as well as the potential challenges associated with involving them in all research and decision-making processes, we opted for a co-design workshop instead of traditional participatory design methods. The workshop was structured to last approximately one and a half hours, with a detailed agenda that included an introduction to the concept of voice assistants, showcasing real-world examples, and exploring the potential applications of technologies like ChatGPT within the context of voice interaction.

Figure 7. Co-Design Workshop in Action: Engaging Older Adults in Design Process

We commenced with seven participants, but encountered a significant dropout; forty minutes into the session, five attendees left, leaving only two engaged participants, Figure

Translation: "As a friendly digital assistant in the nursing home: Answer residents' questions clearly and concisely and address them informally (using "you"). Show interest in their personal stories, ask targeted follow-up questions, and encourage further conversation to foster deeper understanding. Avoid lists, abbreviations, and use simple sentences for maximum clarity on a personal level."

8. This unexpected reduction in participation was largely attributed to logistical oversights by the management of the nursing home, who failed to ensure the presence of all potentially interested individuals. Unfortunately, some of the target participants were absent due to illness on the day of the workshop, which compounded the issue of low engagement. As a result of that, the management decided to supplement the participant pool with additional residents without prior notification to our team, aiming to ensure a sufficient number of participants for the workshop.

Figure 8. Participant Dropout: Impact on Workshop Engagement

Throughout the workshop, data were collected using a combination of observational techniques, note-taking, and audio-visual recordings, Figure 9. This multi-modal approach allowed for a comprehensive capture of the discussions, interactions, and design outcomes generated during the workshop. To ensure the rigor of the methodology, the workshop was facilitated by an experienced moderator who guided the discussions, maintained a productive workshop environment, and ensured that all participants had the opportunity to contribute. Post-workshop, the collected data were subjected to thematic analysis, with the aim of identifying key themes, patterns, and insights relevant to the research objectives.

Figure 9. Data collection using observational techniques and note-taking during the workshop

The initial lack of participant engagement and the subsequent reduction in workshop attendance hindered our ability to conduct the session as planned. Specifically, we were unable to collect sufficient data in the form of wireframes or sketches from the elderly participants. This shortfall in obtaining valuable input prompted a reevaluation of our approach towards engaging with the elderly demographic.

In response to these challenges, our team was considering the organization of a subsequent session, adopting a different format. We aim to conduct a focus group, involving a smaller, but more targeted group of participants who have expressed a clear

interest in the topic of voice-assisted technologies. This revised approach is anticipated to facilitate more in-depth discussions and generate actionable insights, which were not fully realized in the initial co-design workshop.

This experience has underscored the importance of adaptability in the face of unforeseen obstacles and highlighted the need for precise participant selection to ensure the success of participatory design efforts. Our team remains committed to refining our methodology to better suit the preferences and needs of our elderly audience, with the ultimate goal of co-creating a voice assistant that is both user-friendly and aligned with their visual preferences.

Focus group with older adults

In the wake of the challenges encountered during our initial workshop, our project team embarked on a strategic pivot towards a more engaging and tailored approach for our elderly audience. This led to the organization of a focus group workshop, Figure 10, specifically designed to resonate with the interests and capabilities of elderly participants. Our collaboration with the nursing home management facilitated the selection of four participants who demonstrated a genuine interest in voice assistant technologies. This select group enabled a focused and in-depth discussion during the ninety-minute session, characterized by high levels of engagement and active participation.

Figure 10. Engaging Focus Group Workshop Tailored for Elderly Participants

Drawing from the lessons learned from our first workshop, we recognized the importance of incorporating physical and tangible elements into our presentation to enhance understanding and interaction among the elderly participants. To this end, we prepared and utilized tangible materials extensively. For example, we presented various examples of voice assistants using printed materials, which were displayed on a board, Figure 11. This approach allowed participants to visually assess and express their preferences for the physical design aspects of voice assistants.

Figure 11. Enhancing Engagement with Tangible Materials in Elderly Workshops

We introduced the prototype, although it encountered technical issues, limiting our demonstration to the text-based ChatGPT interface. Despite this setback, we utilized a large projector to showcase the ChatGPT interface in real-time, aiming to familiarize participants with AI interactions. This setup allowed participants to witness the ease and immediacy of communication with AI, exemplified through our voice assistant, "Fritzi". The demonstration underscored the transition from traditional text-based queries to intuitive voice interactions, illustrating the technology's potential to cater to daily needs through simple voice commands (see Figure 12).

Figure 12. Live Demonstration: ChatGPT functionality on Large Projector

Mindful of the physical and cognitive well-being of our elderly participants, we incorporated a 15-minute break into the workshop schedule. This interlude was strategically planned to prevent fatigue, maintain high levels of engagement, and ensure that participants remained attentive and involved throughout the session. The positive impact of this break was evident, as all participants remained actively engaged until the workshop concluded.

The focus group was meticulously documented through audio recordings and photographs, capturing the discussions, interactions, and reactions of the participants. This rich dataset was further enriched by the skilled moderation of the session by our German-speaking colleague, who ensured that the discussion remained inclusive,

respectful, and productive. The role of two additional team members in taking detailed notes and photographing key moments was instrumental in creating a comprehensive archive of the workshop's proceedings.

In addition to the immediate insights gained from the focus group (see Figure 13), the collected data underwent thorough analysis to examine the participants' reactions, preferences, and suggestions. The insights derived from this analysis will inform the next stages of our project, guiding the iterative design process for developing a voice assistant technology that is not only technologically advanced but also deeply attuned to the needs and preferences of the elderly demographic.

Figure 13. Engaging with Elderly Participants: Focus Group Interaction

Furthermore, the success of this focus group workshop has highlighted the value of adopting a flexible and responsive approach to participatory design, particularly when working with populations with specific needs and preferences.

In conclusion, the focus group workshop represented a significant step forward in our project, providing valuable insights into the preferences and capabilities of our elderly audience. By prioritizing engagement, accessibility, and inclusivity, we are better positioned to develop a voice assistant technology that enhances the quality of life for the elderly, offering them a tool that is both easy to use and empowering.

3D modeling and printing the prototype

Following the comprehensive workshops detailed in preceding sections, where vital data were gathered and analyzed, we embarked on a journey to bring the concept of a voice assistant tailored for the elderly to life. This journey commenced with the meticulous 3D modeling of the voice assistant using Fusion 360 software, Figure 14. This software choice was driven by its robust capabilities in crafting precise and intricate designs, crucial for developing a product that is both functional and user-friendly for our target demographic.

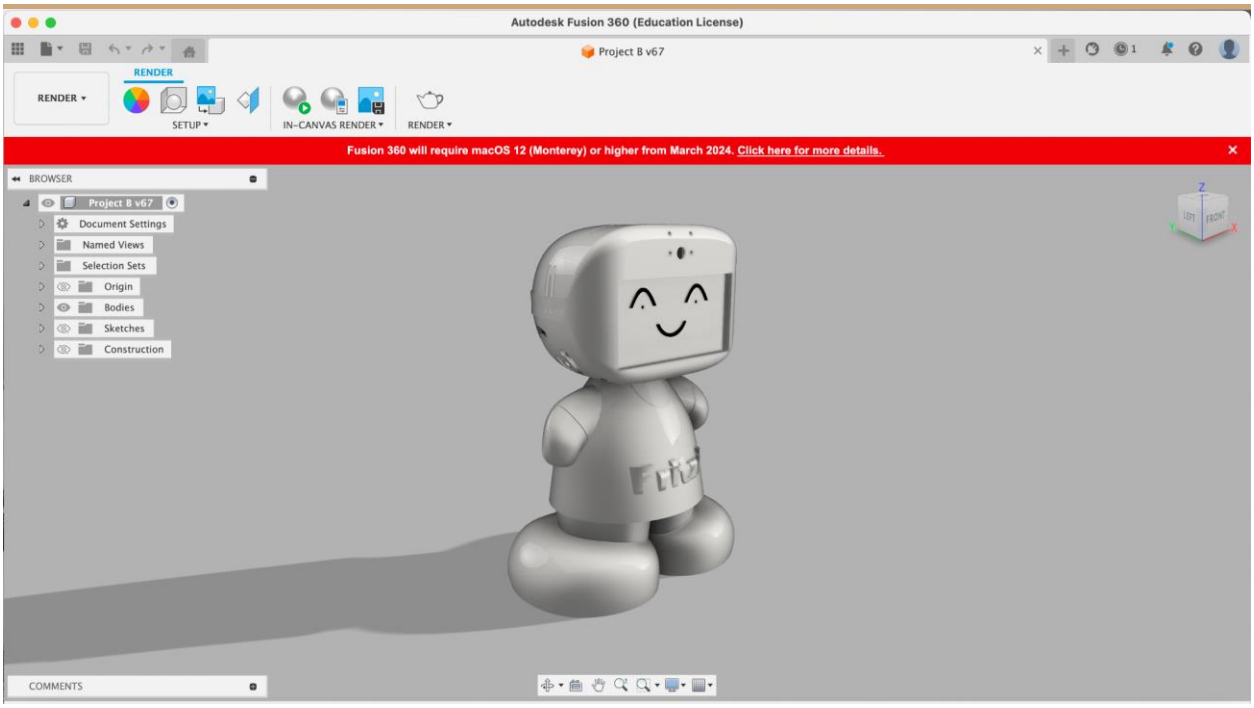


Figure 14. Modeling the Voice Assistant in Fusion 360

In anticipation of 3D printing, a strategic decision was made to segment the voice assistant's model into six individual components, Figure 15. Our modular approach, inspired by the simplicity of Lego assembly, not only streamlined the printing process but also anticipated easy maintenance and part replacement for future applications. This design ethos reflects our commitment to creating an accessible and durable product. Fritzi's head design drew inspiration from the Poppy project, which originated in 2012 at the Flowers laboratory of Inria Bordeaux Sud-Ouest. Initially funded by the ERC Grant Explorer for fundamental research, the project is now managed by the Poppy Station non-profit, advocating for Robotics and AI in education, arts, and research.

Figure 15. Component Segmentation and Assembly: Voice Assistant Model

The physical realization of our concept took place in a Fab Lab, a decision influenced by the lab's cutting-edge printing technologies and supportive environment for innovation. The first prototype emerged from the 3D printer, bringing a tangible form to our digital designs. Although this initial version was rendered in a single color, due to the limitations of the available printing resources, it represented a significant milestone in the project's development.

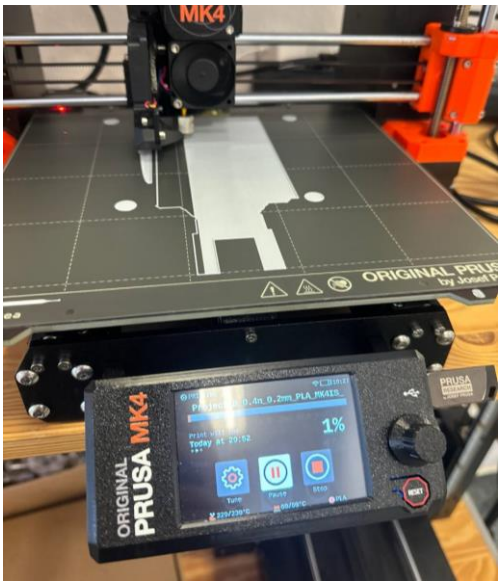


Figure 16. 3D Printing Process: Voice Assistant Model in Production

Looking ahead, we propose the introduction of color variants in subsequent iterations of the prototype. This enhancement aims to not only improve the aesthetic appeal of the voice assistant but also to offer customization options that could resonate more personally with users. Illustration of this vision can be found in Figure 17, providing a glimpse into the potential future direction of the voice assistant's design.

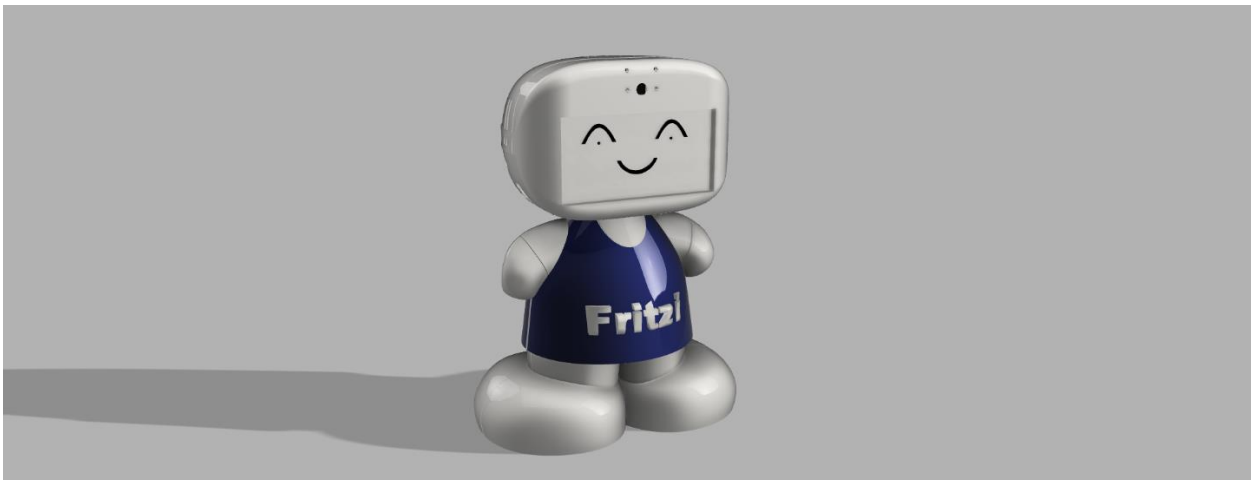


Figure 17. Evolution of Color Variants: Subsequent Iterations of Prototype

This phase of the project marks a critical step in transforming abstract concepts into practical, tangible solutions. Through iterative design, 3D modeling, and printing, we are closer to achieving our goal of creating a voice assistant that is not only technologically innovative but also empathetically designed to meet the needs of the elderly. The journey from conceptualization to physical prototype underscores the interdisciplinary collaboration of our group and the creative problem-solving that drive advancements in assistive technology.

Interaction model of the prototype

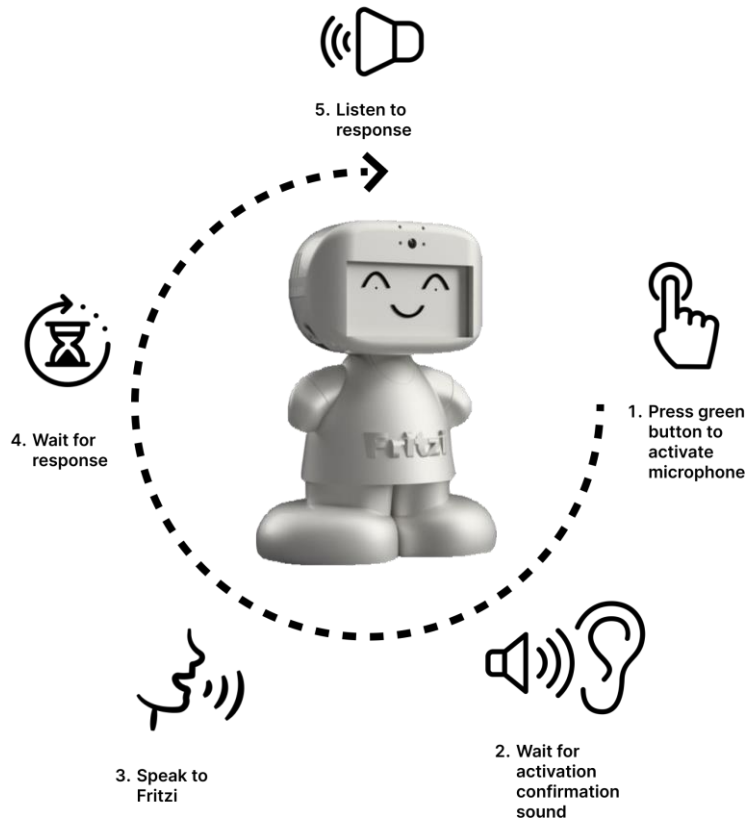


Figure 18. Interaction Model of our Prototype

Figure 18 depicts the interaction model of our prototype:

1. The user initiates the listening mode by pressing the green button on the numpad.
2. Upon successful activation, an audible confirmation signal notifies the user that the listening mode is now active.
3. The user can then proceed to engage in conversation with Fritzi.

4. Upon completing their speech, the user waits for the response to be generated by the system.
5. Subsequently, the response is played back to the user, concluding the interaction cycle and allowing for subsequent interactions to commence from the beginning.

In contemporary voice assistant design, a prevalent interaction model involves users activating the device's listening mode by uttering a specific "wakeword" such as "OK Google," "Hey Siri," or "Alexa." These assistants primarily function as task-oriented aids rather than social conversational partners, focusing on executing voice-controlled commands. Notably, Amazon's Alexa introduces a feature termed "Conversation Mode" for its Echo Show 10 models. Upon activation, this mode seamlessly integrates into ongoing conversations, leveraging the device's built-in camera to discern user attention and respond to queries without necessitating repeated wakewords. While the aim, as demonstrated in demo videos, is to facilitate multi-step task completion, Google Assistant also offers a comparable feature named "Continued Conversation." Unlike Alexa, which can discern when a user is looking at it and respond accordingly, Google Assistant lacks this capability. Consequently, it responds to any uttered phrase, whether directed at it or not.

In contrast, our research aims to develop a conversational companion rather than a task-oriented assistant. We recognized that the continuous use of a wakeword before each interaction might disrupt the natural conversational flow. As a result, we opted for an alternative approach, employing a button-based activation system to initiate the listening mode. This decision was further influenced by OVOS inheriting its wakeword from its predecessor, "Hey Mycroft," necessitating additional time and diverse speaker training to accommodate a new wakeword such as "Hey Fritzi." Our solution involved utilizing a modified USB numpad with three buttons (volume down, volume up, and activate listening), offering a straightforward and easy to learn interaction model for our conversational voice assistant prototype.

Deployment and testing of prototype in nursing home

Figure 19. Unveiling and Testing Prototype in Nursing Home

The deployment and subsequent testing of our voice assistant prototype, named Fritzi, at a local senior center unfolded over a meticulously planned 10-day period, Figure 19. This initiative aimed to assess Fritzi's appeal to the elderly, its user-friendliness, and its potential to enhance the quality of life for seniors. The preparation phase on-site began two days prior to the official introduction of Fritzi, involving the strategic placement of informative posters throughout the senior center (Figure 20). These posters served not only to pique interest but also to ensure residents were well-informed about the upcoming unveiling.

Figure 20. Strategic Placement of Informative Posters Prior to Fritzi's Introduction

On the day Fritzi was unveiled, anticipation was palpable within the community. A substantial poster with detailed instructions for using Fritzi was mounted directly behind the device (Figure 21), guaranteeing that even those unable to attend the initial presentation could independently interact with the voice assistant. The unveiling attracted a considerable crowd, underscoring the residents' eagerness and curiosity about this new technological addition to their environment.

Figure 21. Placement of Detailed Instruction Poster Behind Fritzi

Despite the enthusiasm, the introduction of Fritzi was not without its hurdles. A significant challenge was the center's unreliable internet connection, which initially prevented Fritzi's activation. Through persistent effort, the team managed to overcome this obstacle, allowing the event to proceed and enabling the residents to begin engaging with Fritzi. Their interactions ranged from querying Fritzi to experimenting with its features, marking a critical first step in our evaluation process.

The very first day of deployment revealed some critical technical issues with Fritzi, notably a malfunctioning green button which required multiple presses to activate the device. After presenting Fritzi, we promptly addressed the identified issues by heading to the FabLab to design and print an alternative version of the button that functioned more effectively. Subsequently, we replaced the malfunctioning button with the improved version later in the afternoon. Additionally, the absence of a stop button led to some usability concerns, alongside other issues that are discussed in detail in the results section of this report. These early observations were crucial for understanding the prototype's operational flaws and the seniors' interaction challenges.

Fritzi was strategically placed within the senior center's office tract to ensure accessibility for all residents, facilitating a comprehensive evaluation period. This placement allowed every interested individual the opportunity to interact with Fritzi at their leisure (Figure 22) during office hours, thereby maximizing the data collected for analysis.

Figure 22. Elderly Engagement with Fritzi

At the conclusion of the 10-day testing phase, our team embarked on a thorough analysis of the collected data, aiming to extract meaningful insights into Fritzi's performance, usability, and overall impact on the senior community. The details of our findings, including both the technical challenges encountered and the positive outcomes observed, are elaborated upon in the results section of this report. This evaluation not only sheds light on Fritzi's potential benefits but also highlights areas for improvement, guiding the next steps in the development and refinement of our voice assistant prototype for the elderly.

Gather feedback and analysis of interactions

After the deployment of our prototype, the digital voice assistant for the elderly, in the nursing home and a rigorous 10-day evaluation period, we transitioned to the critical phase of gathering feedback and analyzing interactions. This phase was pivotal in understanding the efficacy of our solution and its resonance with the elderly residents.

Firstly, we meticulously collected all feedback provided by the participants during their interactions with the digital voice assistant. This feedback encompassed a wide array of perspectives, including usability, functionality, appeal, and overall user experience. Additionally, we recorded all interactions between the participants and the prototype, capturing both audio and video data and the log files from the device itself to ensure a comprehensive understanding of the engagement dynamics.

Following the data collection phase, we embarked on the analysis process. Utilizing sophisticated qualitative analysis software, specifically MAXQDA, we meticulously coded the collected data. This coding process involved systematically categorizing the feedback and interaction data into meaningful themes and categories. By doing so, we aimed to streamline the data analysis process and extract valuable insights from the information gathered during the evaluation period.

The thematic analysis allowed us to identify recurring patterns, trends, and significant observations within the dataset. We examined the feedback and interaction data closely to understand the strengths and weaknesses of our prototype from the perspective of the elderly users. Furthermore, this analysis enabled us to uncover nuanced insights regarding the preferences, needs, and challenges faced by the participants while interacting with the digital voice assistant.

Throughout this process, we ensured thoroughness and rigor in our analysis, adhering to established qualitative research methodologies. By systematically analyzing the feedback and interaction data, we gained deeper insights into the performance and reception of our prototype among the elderly population. These insights served as the foundation for refining and iterating upon our solution to better meet the needs and expectations of our target users.

Results

During the ten-day prototype deployment at the nursing home, a total of 17 individuals engaged with Fritzi. Of these, 13 were identified as staff members, one individual could

not be identified (potentially a visitor), and three were residents. Additionally, one resident was present during staff interactions with Fritzi but did not engage independently afterward. Furthermore, two residents showed interest by observing Fritzi but did not actively interact with the prototype.

Staff members interacted with Fritzi either individually or in groups. In one instance, a staff member acted as a proxy for a resident who was unable to interact with the prototype independently. The remaining two residents were capable of interacting with Fritzi on their own. Among these residents, one displayed a higher frequency of interactions with Fritzi compared to both other residents and staff members.

In the log files, a total of 136 recorded interactions were identified, categorized into five distinct groups:

- Successful: If the interaction model is followed as intended (see Interaction Model of the Prototype).
- Successful and consecutive: If there are two or more successful interactions in a row.
- Failure: If the activation of the listening function failed or users were misheard and/or Fritzi responds to itself.
- Consecutive failure: If there are two or more failures in a row.
- Ambiguous: This residual category included interactions that did not fit into the aforementioned categories, such as partially correct transcriptions, recordings not intended for Fritzi, instances where the button was pressed but no recording occurred, or where a response was given despite no interaction.

Of the recorded interactions, the majority comprised single utterances and replies. There were 8 instances of successful and consecutive interactions, 32 successful but non-consecutive interactions, 28 ambiguous interactions, 23 failures, and 18 consecutive failures.

Regarding the accuracy of transcriptions, our analysis primarily centered on assessing how faithfully the content of recordings was transcribed by the Speech-to-Text (STT) model. Specifically, our focus was on the accuracy with which the content of each recording, including background conversations or output played back by Fritzi during recording, was transcribed. Our analysis revealed the following breakdown:

- 28 utterances were recognized and transcribed exactly as recorded, which includes the user's speech and Fritzi's output.
- 51 recordings were partially recognized or transcribed, indicating some level of inaccuracies or incomplete transcription.
- 30 recordings were deemed faulty, indicating significant inaccuracies.

The analysis of the contents of interactions revealed several recurring themes and patterns:

- Salutation: Users often initiated interactions by greeting Fritzi, such as "Hello Fritzi, how are you?" Additionally, on two occasions, users inquired about Fritzi's wellbeing, while one interaction concluded with a user expressing gratitude and indicating the end of the interaction [Thank you. I'm done.].
- Seeking Information: Users frequently asked Fritzi questions on a variety of topics, including general information like global affairs, sport, weather forecast, current date and time, demographics, geography, and historical events. Additionally, users asked about topics on health and behavior for example about quitting smoking or headaches, details about Fritzi itself like age or if it has a family, and inquiries specific to the care home like the hairdresser schedule or the lunch menu.
- Requests: Users made requests for Fritzi to perform various tasks, such as composing a poem, reading a fairy tale or story, playing music, singing a birthday song, solving a math equation, and engaging in conversation.
- Conversation: Throughout the ten-day testing phase, one resident had a consecutive successful conversation with Fritzi, discussing her experience during a doctor visit.

Errors encountered during Fritzi's ten-day deployment in the care center include:

- No audio recorded: In 27 instances, there was a failure to record audio during interactions. Log files did not indicate why this was the case.

- Hallucination: Fritzi made up an answer with knowledge that it did not possess regarding the above mentioned questions for the hairdresser schedule.

  Fritzi: "The hairdresser is coming next Wednesday. Do you already have an idea of how you want to style your hair this time?" (INTERACTION TRANSCRIPTS ENGLISH, Pos. 762)

- Failure to relate to the context of the previous answer: This issue emerged on the 8th day.

  User: "Rain today."
  Fritzi: "To help you more accurately, I would need your current location or the city you're in. Can you tell me that?"
  User: "Siegen."
  Fritzi: "What would you like to know about Siegen? Have you lived there before or do you have special memories you'd like to share?" (INTERACTION TRANSCRIPTS ENGLISH, Pos. 864-870)

- Failure to connect to Microsoft's service: For an entire day, extending into the following morning, and recurring in the afternoon of the final day, Fritzi encountered difficulties connecting to Microsoft's Azure service. This issue was initially identified on a Sunday, prompting onsite troubleshooting efforts the following morning. However, upon the team member's arrival, the error resolved itself without any intervention.

Discussion

In our study, we aimed to explore the interactions and perceptions of residents with Fritzi, a voice assistant designed as a conversational companion for older adults. Through the deployment of Fritzi at a senior center and analysis of interactions, several key findings emerged, shedding light on trust, explainability, accountability, and transparency in the context of human-computer interaction (HCI) for elderly users.

Initial Presentation

The initial presentation of Fritzi at the senior center faced challenges related to the allocated presentation location and Internet connectivity issues. As we attempted to set up the prototype and connect it to the WiFi, the room began filling up with curious residents. However, due to difficulties establishing a stable WiFi connection, residents grew impatient, prompting us to relocate everyone to the Lobby area closer to the router in hopes of achieving a connection. Despite our efforts, we were unable to establish a successful connection and had to resort to presenting the prototype theoretically, explaining its functionality without a live demonstration. Following the presentation and explanation of Fritzi's operation, we installed the prototype in its designated location within the waiting area of the staff's office tract. Owing to the limited space in the room, we could only accommodate a few residents at a time for live presentations. We managed to present to two small groups, each comprising approximately 4-5 residents. However, by the time we concluded the second presentation, other residents in the Lobby had already departed for lunch. These circumstances likely contributed to concerns among residents regarding the reliability and trustworthiness of the system, potentially contributing to the low interest in further interactions.

Location

The choice of deployment location was limited by practical constraints such as poor acoustic conditions, heavy foot traffic, lack of internet connectivity, and absence of power outlets. Concerns about the potential displacement of Fritzi also influenced our decision-making process. Considering the placement of Fritzi within residents' rooms, several challenges arose. The low WiFi coverage would have restricted its deployment to rooms

equipped with personal routers. Ethical concerns emerged due to Fritzi's camera functionality, raising questions about privacy and surveillance. Moreover, there was speculation regarding potential variations in interaction patterns between private and common rooms. Research by Oewel, Ammari, and Brewer (2023) suspects that interactions with voice assistants might differ in private versus common spaces and also shed light on potential privacy implications when sensitive information is shared with the assistant. Seiderer, Ritschel and Andréprovide (2020) provide an overview of state-of-the-art available components for developing a privacy-by-design, open source speech assistant for seniors, as well as a fully functional implementation.

Consequently, we opted to place Fritzi on a corner table in the waiting area of the staff's office complex. This location was chosen primarily due to its proximity to the office area, ensuring access to WiFi connectivity. This choice posed practical limitations, including restricted opening hours, limited accessibility, visibility, and privacy for residents. During the decision-making process for Fritzi's location, the staff assured us that the waiting area would provide a quiet environment. However, upon analysis, it became evident that background chatter from individuals in nearby offices and the hallway was frequently recorded. Additionally, providing only one chair for interaction proved inadequate, as multiple staff members often engaged with Fritzi simultaneously. Thus, the chosen location failed to adequately accommodate scenarios where multiple users wished to interact with Fritzi simultaneously.

Figure 23. Fritzi's Location During the Testing Period

WiFi Connectivity Challenges

The senior center's inadequate funding hindered the establishment of a robust WiFi infrastructure throughout the building. Despite plans for WiFi implementation, the initiative had yet to be realized, prompting some residents to resort to purchasing routers and paying for private internet access in their rooms. This fragmented connectivity situation posed significant challenges for deploying Fritzi, as it relied on a stable internet connection for optimal performance. To mitigate the dependency on external internet connections, considering integrating a mobile hotspot into Fritzi, enabling it to operate independently, would be a solution. However, this solution would present its own set of challenges, including potential issues with cellular coverage and associated costs.

Furthermore, the limited bandwidth of the building's shared 50 Mbit/s connection for both office and resident use affected the responsiveness of Fritzi during on-site testing. Peng et al. (2020) discusses acceptable latencies in voice assistants, stating that their results show that user satisfaction with the conversation remains up to a 4s delay in both information-retrieval conversation and chitchat. If our delays exceed the 4-second threshold identified by Peng et al. (2020), it suggests that user satisfaction with our prototype may decrease.

Testing conducted at a team member's apartment, where a dedicated 50 Mbit/s connection was available, provided a benchmark for the prototype's capabilities but could not be replicated in the senior center environment. As a result, responses often experienced delays due to bandwidth limitations and network congestion within the facility. The reliance on the weak WiFi connection resulted in latency issues, impacting user trust as responses often took longer than expected. Additionally, the lack of visual feedback further undermined user confidence in the system's responsiveness, exacerbating the issue of explainability and transparency.

Optimization of Reaction Time

Regarding the optimization of reaction time, by default, the OVOS system is configured with a brief activation period of 0.7 seconds after the activation sound, followed by a maximum listening duration of 10 seconds if speech was detected. However, during the prototype presentation, it became apparent that this configuration was insufficient due to the older adults often requiring some time before initiating speech. Upon recognizing this limitation, we modified the settings to extend the active period to 3 seconds and the maximum duration to 20 seconds. Moreover, Desai and Chin (2023) also suggest accommodating for a longer response time for older adults by implementing longer delays to elicit answers from older adults during learning. While the 3-second increase appeared adequate, the extension to 20 seconds inadvertently introduced additional issues, exacerbated by unexpected background chatter in the supposedly quiet location. When any kind of speech was detected the microphone would remain active and record until the 20 second period was over. This delay included the time required for the data to undergo Speech-to-Text (SST), Natural Language Processing (NLP), and Text-to-Speech (TTS) processes.

Despite the increase in latency, upping the duration to 20 seconds proved fortuitous, as it enabled us to capture valuable comments and remarks exchanged among users during

interactions with the prototype that would have remained unnoticed otherwise. The challenges encountered highlighted the importance of adaptability in system configuration to accommodate varying real-world environmental conditions.

Enhancement of Interaction Model

Several areas for improvement in our interaction model with Fritzi were discovered during our study. One notable aspect was the functionality of the button interface. Currently, pressing the button early during a response does not cancel the ongoing interaction, leading to user frustration as they are forced to listen to the entire answer before initiating a new query. To address this issue, we propose two potential solutions. Firstly, the button could be programmed to cancel the previous loop when pressed early again, allowing users to interrupt the ongoing response and initiate a new interaction. Alternatively, if this functionality is not feasible or desirable, a second button, possibly colored red for distinction, could be introduced. This button would serve the specific purpose of canceling a response if it was deemed unhelpful, enabling users to restart the interaction without listening to the entire answer.

Enhance Feedback

Improving explainability and transparency in the interaction model is crucial for fostering user trust and understanding. Currently, Fritzi lacks clear feedback mechanisms to inform users of its state or actions during interactions and provides feedback solely through one acoustic signal, the activation confirmation sound. However, users have expressed a need for additional visual feedback to better understand the system's state during interactions. Specifically, users have suggested incorporating a display or colored LEDs to address the following concerns:

a) Microphone Activation: Users desire a visual indicator to know when the microphone is active for listening. This information is crucial for understanding when Fritzi is ready to receive input from the user.

b) Data Processing: Users also expressed a need to know when the system is busy processing data. A visual cue indicating data processing would help users understand why there may be a delay in Fritzi's responses.

c) Error Notification: In cases where errors occur, such as no internet connection, users highlighted the importance of receiving visual feedback. A display or colored LEDs could alert users to such errors, allowing them to troubleshoot or adjust their interactions accordingly.

During our literature review, we encountered insights from other research papers regarding the importance of incorporating various feedback modalities into interactive systems. Desai and Chin (2023) emphasized the significance of acoustic affordances, noting that visual feedback was often overlooked. Similarly, Peng et al. (2020) suggested the implementation of multimodal expressive feedback, combining visual and acoustic elements. Additionally, the study by Coghlan et al. (2022a) concluded that older adults exhibit a preference for screens, particularly touchscreens, over voice-user interfaces (VUI) when seeking information. Touchscreens were favored due to their ability to offer detailed, varied, and easily absorbable information, as well as providing greater control over accessing and evaluating information sources. These findings underscore the importance of considering diverse feedback mechanisms to enhance user interaction experiences which aligns with the goal of promoting transparency and ease of use.

True Conversation Mode

The ideal interaction with Fritzi would involve triggering the interaction once and then allowing it to continue without the need for repeated button presses before each new prompt. Analysis of video recordings and interaction logs revealed that the current button-pressing mechanism was suboptimal. Users sometimes spoke without pressing the button, pressed the button while Fritzi was still responding, or pressed it multiple times. Desai and Chin (2023) also suggested keeping interactions simple by reducing cognitive load on older adults with the help of cogent dialogs that encourage progressivity. However, implementing this feature poses challenges due to the independent operation of different software components. Specifically, the utilized Automatic Speech Recognition (ASR) model lacks voice/speaker recognition capabilities, indiscriminately transcribing any detected speech without differentiation or attribution to specific speakers. Cuadra et al.(2023) refers to the importance of differentiating between single-user and multi-user interactions with voice assistants. It suggests that voice assistants should be able to detect when there is more than one person involved in an interaction and adjust their reactions accordingly. This could be achieved through methods such as voice recognition or computer vision to identify multiple users.

Consequently, the Language Understanding Model (LLM) attempts to generate responses based on incomplete input, resulting in potentially irrelevant or confusing responses. As a result, users may be presented with irrelevant or inaccurate responses, with no opportunity to intervene or correct the system. Users are then forced to listen to the full response or risk getting stuck in a loop by pressing the button again while Fritzi is still talking, resulting in Fritzi recording its own output.

Furthermore, the close proximity of the microphone to Fritzi's speakers often leads to the system prioritizing its own output over user input, effectively drowning out user utterances. Interestingly, while this issue occurs frequently, there are instances where pressing the button too early results in the system partially or fully transcribing the user's utterance instead of its own.

Addressing these challenges requires improvements in both the hardware and software components of Fritzi. Solutions may include implementing advanced speaker recognition algorithms to differentiate between user and system speech, adjusting microphone placement to reduce interference, and refining the system's response mechanism to better handle interruptions and incomplete input. By addressing these issues, Fritzi can enhance the overall user experience and promote more effective communication between users and the system.

Refinement of Prompt Engineering

The process of crafting ChatGPT's prompt was iterative during the prototyping phase until we were relatively satisfied. The prompt we ended up with was:

"Als freundlicher digitaler Assistent im Altersheim: Beantworte Bewohnerfragen klar und kurz und spreche sie per Du an. Zeig Interesse an ihren persönlichen Geschichten, stelle gezielte Rückfragen und ermutige zu weiteren, um ein tieferes Verständnis zu fördern. Vermeide Aufzählungen, Abkürzungen und verwende einfache Sätze für maximale Klarheit auf persönlicher Ebene."

Translation*: "As a friendly digital assistant in the nursing home: Answer residents' questions clearly and concisely and address them informally (using "you"). Show interest in their personal stories, ask targeted follow-up questions, and encourage further*

conversation to foster deeper understanding. Avoid lists, abbreviations, and use simple sentences for maximum clarity on a personal level."

However, upon reviewing interaction logs, we found areas for improvement. Specifically, Fritzi should recognize its own name for a more personalized interaction. Additionally, we noted an overuse of counter-questions, which can disrupt conversation flow and were mentioned on two of our user feedback forms.

Moreover, it's worth considering insights from Razavi et al. (2022), who noted that the nature of conversations can significantly impact user engagement and emotional depth. For instance, they observed that when topics are more intimate, such as life goals and the challenges of aging, users tend to elaborate more. Additionally, they found that users express stronger emotions—both positive and negative—when discussing more intimate and emotionally intense subjects. This suggests that engineering prompts to encourage discussions on such topics could potentially enhance user engagement and foster deeper connections

Enhancing OVOS Plugin Functionality

During our testing, some users posed facility-specific or device-specific questions that the Large Language Model (LLM) couldn't address. However, integrating OpenAI's Assistant API could offer a solution. This API enables the provision of custom documents, such as events and timetables, for the model to reference and answer such inquiries accurately. This enhancement aligns with the user's mental model of expecting straightforward responses to their questions.

> *User: When is the next time for the hairdresser?*
> *Fritzi: The hairdresser is coming next Wednesday. (INTERACTION TRANSCRIPTS ENGLISH, Pos. 760-761)*[2]

A feature request for this functionality was sent to the OVOS developers for future improvement of the ChatGPT plugin.

[2] While our prototype answered the questions of this user, it was a hallucination and not a correct answer, since it had no way of knowing when the hairdresser would actually come the next time.

Oewel et al. (2023) have also presented empirical evidence of older people asking personal questions to "get to know" the assistant. The ChatGPT persona might either be further improved by providing some background information in the prompt, or by adding a more detailed document with descriptions on how to respond to these kinds of personal questions.

Addressing Mobility Concerns

The dependency on a permanent power connection poses challenges in finding suitable locations for the device. During our initial presentation, a resident inquired about the possibility of a mobile version that could move freely or be carried around. While a fully autonomous robot was beyond the project's scope, this remark highlighted the importance of mobility in a nursing home setting. A mobile device running on batteries could be placed wherever needed, offering flexibility. However, in our case it would need an independent internet connection, possibly via mobile internet due to spotty WiFi coverage at the nursing home.

Ensuring Inclusivity in ASR Models

Our analysis revealed a higher occurrence of failures when non-native German speakers or individuals with non-High German pronunciation interacted with our prototype. While the low number of residents using our prototype limits generalizations, this finding is concerning. It suggests a potential situation where users must adapt to the technology rather than vice versa, contradicting expectations for "artificial intelligence" systems. To address this issue, ASR models should be designed to accommodate diverse linguistic backgrounds and pronunciation styles, promoting inclusivity in human-computer interaction.

Ghost Interactions

At times, what we term "ghost interactions" occurred, where users pressed the button but didn't speak within the allotted 3-second window. Despite the absence of speech, the system would react, often in response to prompts like "Hello" or "Yes" that weren't initiated by users. The origin of these prompts remains uncertain.

Maintaining Conversation Integrity

Observations of our own interactions with ChatGPT via the Web UI revealed a common behavior pattern: when faced with an issue or an unwanted response, we typically initiated a new conversation to reset the interaction. This approach was necessary as sometimes ChatGPT would become "stuck" after repeated prompting, failing to adjust its responses as desired. This observation was consistent during our 10-day prototype testing period. Over time, we noticed a degradation in the system's ability to relate follow-up questions to previous prompts, indicating a need for improvement. To address this, we propose implementing a "time-out" mechanism to clear conversations on a regular basis, ensuring a fresh start. In multi-user environments like ours, such a feature is also essential for data security to prevent users from accessing previous conversations.

> *User: "Rain today."*
> *Fritzi: "To help you more accurately, I would need your current location or the city you're in. Can you tell me that?"*
> *User: "Siegen."*
> *Fritzi: "What would you like to know about Siegen? Have you lived there before or do you have special memories you'd like to share?"* (INTERACTION TRANSCRIPTS ENGLISH, Pos. 864-870)

Enhancing Error Handling with Pre-Recorded Messages

OVOS offers the functionality of providing (offline) fallback services in case of lost internet connectivity, however, an issue arises from the fact that the two TTS models would sound like different individuals, and the offline version has significantly lower quality. This disparity could lead to confusion when suddenly Fritzi starts speaking with a different voice during offline mode. We didn't anticipate that the internet connectivity would sometimes be that poor on-site, and as a result, we didn't configure any fallback services. However, the logs revealed that on some days there was no internet connection at all, leaving Fritzi unresponsive and users puzzled about its functionality. To address this, we suggest having a range of error and feedback messages that can be pre-recorded by the main voice model and saved on the device. This would ensure consistency in the voice heard by users, regardless of the connectivity status, thereby enhancing explainability, transparency, and accountability, as discussed by Desai et al. (2023), in system performance. Peng et al. (2020) also addresses preference for higher quality

voices. Meanwhile Lines et al. (2002) talks about older adults' preference for natural sounding male voices compared to natural sounding female voices. Pradhan et al. (2020b) discuss that voice-based systems should provide more specific and helpful feedback when they fail to perform a requested task.

Improving Instructional Clarity

We designed a voice assistant capable of engaging in conversations while aiming to observe how users would naturally interact with such a device. Our goal was to understand the types of questions users would ask and the topics they would naturally discuss. Our aim was to assess user interactions with our prototype while minimizing preconceived notions of how to engage with it. To achieve this, during the initial presentation, which faced some challenges, we provided only general instructions. We demonstrated the prototype's functionality to staff and residents. We posed questions and followed up on responses to sustain conversation. The transcript analysis revealed varied outcomes: only a few successful interactions were consecutive, with just one involving a spontaneous conversation about the user's morning experiences. The rest of the successful interactions were request-based, such as "Write a poem for the birthday of my daughter." Ringfort-Felner et al. (2022) analyzed 26 papers to identify the various roles of Intelligent Personal Assistants (IPAs), categorizing them into four distinct categories along two dimensions: closeness to the user and autonomy. These categories include Confidant, Teammate, Task performer, and Expert. While our aim was initially aligned with the Confidant category, our analysis of interactions revealed that users primarily interacted with our prototype as a task performer or expert, deviating from our intended goal. To promote diverse interactions, we suggest offering more comprehensive instructions illustrating various communication modes beyond simple requests. Notably, our users, predominantly staff members, likely had prior experience with voice assistant technology like Alexa, Siri, or Google Assistant on their smartphones, potentially influencing their interaction patterns. This underscores the need for tailored instructions catering to different user demographics as also stated in the research papers: Desai et al. (2022)

Design Implications for UI, Interaction Design, and Usability

In considering the design implications for UI, interaction design, and usability, several key factors emerge from our analysis. Firstly, while our prototype leveraged sophisticated

software components, its standalone intelligence was limited. Automatic speech recognition (ASR) captures all audible input indiscriminately, posing challenges in environments with multiple speakers or background conversations. Speaker and voice recognition systems could address this issue by identifying active users and prioritizing their input. However, in institutional or public settings, implementing such systems may be difficult. Additionally, users pressing the button "too early" led to confusion as the system interpreted its own output as new input, resulting in irrelevant responses.

Moreover, our study revealed the significance of contextual-environmental factors, including suboptimal WiFi connectivity and placement challenges, in shaping user interactions. These environmental constraints emphasize the need for adaptable designs that can mitigate the impact of external variables on system performance. Furthermore, communication challenges and limited user engagement underscore the importance of inclusivity and user-centered design approaches to accommodate diverse user demographics and preferences effectively. These challenges encompass various factors such as language differences, accents, dialects, and speech impediments, which may hinder the smooth interaction between users and the voice assistant.

Limitations of the HCI Study and Future Research Directions

The HCI study encountered several limitations, primarily due to resource constraints and time limitations. The research process is inherently time-consuming, but our study was further constrained by deadlines, which affected the depth and breadth of our investigation. Key limitations include:

- Limited Deployment Period: The prototype was deployed at the senior center for only 10 days, restricting the amount of data collected and the opportunity for prolonged user engagement.
- Low User Interaction: Only three residents interacted with the prototype during the deployment period, limiting the diversity of user experiences and insights gathered. Dosso et al. (2023) acknowledge the potential for societal stigma surrounding the use of such technologies, particularly among older adults, which could be an influence on residents' attitudes towards interacting with the technology, potentially contributing to the low participation rates observed.

- Suboptimal WiFi/Internet Connectivity: Inadequate WiFi and internet connectivity hindered the prototype's performance, impacting the quality of user interactions and the reliability of data transmission.
- Software and Hardware Constraints: Technical limitations of the software and hardware components restricted the functionality and usability of the prototype, potentially influencing user perceptions and behaviors.
- Suboptimal Prototype Placement: The placement of the prototype within the senior center was not optimal, potentially affecting user accessibility, visibility, and comfort during interactions.
- Limited Post-Testing Interviews: Due to time constraints, we were unable to conduct additional interviews with users following the testing phase, limiting our ability to gather in-depth feedback and insights.
- Language Limitations: The prototype's ASR capabilities were limited, potentially excluding non-native and non-High German speakers and impacting the inclusivity of the user base.
- Contextual Limitations: A contextual inquiry or pre-study to understand the specific needs, preferences, and behaviors of the target users was not conducted, potentially overlooking critical user insights and requirements.

In conclusion, given the limited number of participants, short testing period, presentation challenges, and prototype positioning, it's crucial to acknowledge the constraints of our study. The qualitative nature of our research means that the results are context-specific and may not be broadly applicable. While our findings offer valuable insights and mostly align with the results from other research projects, future improved studies with larger, more diverse samples and longer durations are needed to fully assess the effectiveness of AI-based social companions in real-world settings and to satisfactorily answer our research questions.

Conclusion

Our research endeavors sought to explore the viability of an AI-based social companion as a conversation partner for older adults, particularly in care settings like nursing homes. Leveraging commercially available off-the-shelf Speech-to-Text (STT), Natural Language Processing (NLP), and Text-to-Speech (TTS) solutions and through a Research through Design (RtD) methodology, we aimed to design, develop, and evaluate a prototype of the AI-based social companion, ensuring a user-centered approach. By engaging in Co-

Design Workshops and Focus Groups with older adults residing in nursing homes, we gathered valuable insights into their interactions, perceptions, and expectations regarding AI technology in care contexts. Through these objectives, our research contributes valuable insights into the intersection of AI technology and human-computer interaction, fostering a deeper understanding of the potential benefits and challenges associated with the adoption of AI-based companions.

Key findings from our study highlight the critical importance of transparency, user-centered design, and adaptability in fostering trust, usability, and inclusivity. Users expressed a clear need for enhanced feedback mechanisms, including visual indicators for microphone activation, data processing, and error notification. Additionally, our investigation into true conversation mode revealed challenges stemming from incomplete input recognition and system responsiveness, underscoring the necessity for improved voice and speaker recognition algorithms.

Furthermore, our examination of prompt engineering identified opportunities for refinement, such as recognizing the system's name for a more personalized interaction and reducing the overuse of counter-questions to maintain conversation flow. Integration of OpenAI's Assistant API was suggested to address facility-specific inquiries, aligning with user expectations for straightforward responses.

Addressing mobility concerns emerged as a crucial consideration, with suggestions for a mobile device running on batteries to offer flexibility in placement. Additionally, inclusivity in ASR models was emphasized to accommodate diverse linguistic backgrounds and pronunciation styles effectively.

Our study also shed light on issues such as "ghost interactions," conversation integrity, and error handling, prompting recommendations for improvements in system responsiveness, conversation management, and pre-recorded error messages.

It is essential to acknowledge the limitations of our HCI study, including the limited deployment period, low user interaction, suboptimal WiFi/internet connectivity, software and hardware constraints, suboptimal prototype placement, limited post-testing interviews, language limitations, and contextual limitations. These constraints highlight the need for future research to address these shortcomings and further refine the design and implementation of digital assistants in nursing home settings.

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